Kydland-Prescott model though the lens of Stokey (1989) and Quant Econ lectures 97, 98

Mariia Elkina
University of Pennsylvania

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Motivation

- What outcomes are attainable in a model where the government has an incentive to generate surprise inflation in order to stimulate the economy?
- One-period economy:
  - high inflation without any benefits for stimulating the economy
- Infinite horizon economy:
  - infinite repetition of one-period economy equilibrium is also possible
  - there are better equilibria with lower inflation
- Implementing low inflation equilibria requires history-dependence...
- ... where it's understood that surprise inflation will result in high inflation expectations

Outline

- Environment
- One-period economy
  - Nash equilibrium
  - Ramsey outcome
- Infinite horizon economy
  - SPE definition using histories
  - Recursive representation
  - Abreu, Pearce, and Stacchetti method of finding equilibrium values
  - Worst equilibrium
  - Best equilibrium
  - Other equilibria

Environment

- Time is discrete, $t = 0, 1, ...$
- Agents: continuum of identical private agents, the (benevolent) government
- Social welfare

$$-\frac{1}{2}\delta^t(U_t^2 + y_t^2)$$

where U_t is unemployment, y_t is inflation

- Unemployment is determined by the Phillips curve

$$U_t = \underbrace{U^*}_{\text{natural unemployment}>0} - \theta \underbrace{(y_t - x_t)}_{\text{surprise inflation}}$$

Timing within period t:

- Households form inflation expectations, x_t
- Government chooses inflation, y_t

Equilibrium concept - subgame perfect equilibrium

Rational expectations

- Private agents' expectations depend on what they expect government to do

$$x_t = \mathbb{E}_t(y_t)$$

- Households cannot be surprised in equilibrium
- Rewrite per period social welfare as

$$r(x, y) = -\frac{1}{2}\left[(U^* - \theta(y - x))^2 + y^2\right]$$

One-period economy

- Suppose economy lasts 1 period
- Nash equilibrium is the pair $(\tilde{x}, \tilde{y}) \in \mathbb{R}^2$ such that
  - given $\tilde{x}$, government maximizes social welfare, i.e. $\tilde{y} \in \arg\max_y r(\tilde{x}, y)$
  - private agents' expectations are rational, i.e. $\tilde{x} = \tilde{y}$
- In equilibrium government solves

$$\max_y -\frac{1}{2}\left[(U^* - \theta(y - x))^2 + y^2\right]$$

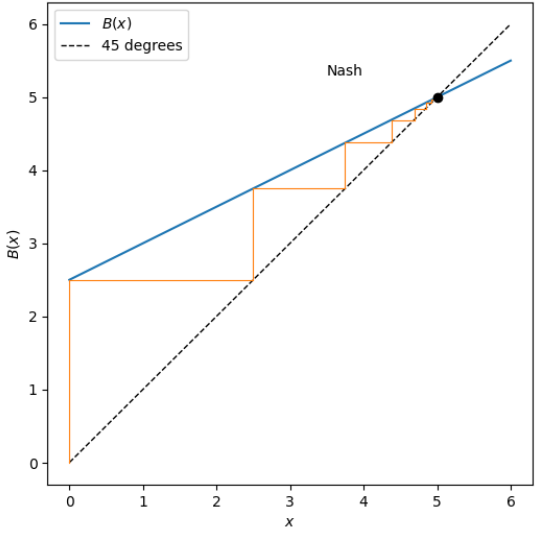
- Government's best response is

$$B(x) = \frac{\theta}{\theta^2 + 1} U^* + \frac{\theta^2}{\theta^2 + 1} x$$

- In equilibrium

$$y = B(y) \Rightarrow x^N = y^N = \theta U^*, U^N = U^*$$

Graphical illustration



Ramsey outcome

- Ramsey outcome solves

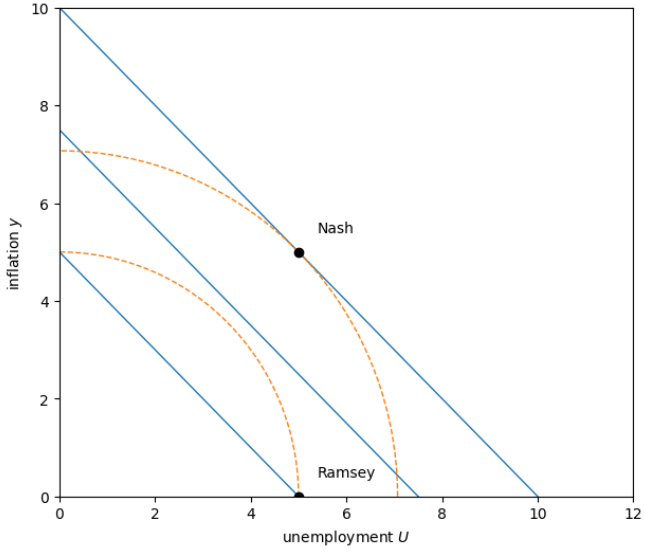
$$\begin{aligned} \max_{x,y} \quad & -\frac{1}{2}\left[(U^* - \theta(y-x))^2 + y^2\right] \\ \text{s.t.} \quad & x = y \end{aligned}$$

- It implies

$$y^R = 0, U^R = U^*$$

- It corresponds to the government committing to policy before households act
- If the government cannot commit, it wants to reoptimize at the end of the period
- Time-inconsistency problem

Graphical illustration



Infinite horizon economy

- Impose inflation bound, $y \in [0, y^{\#}] \equiv Y$
- Define history at t as $h^t = (x_0, y_0, ..., x_t, y_t)$
- Private agents' strategy is $\gamma^X : H^t \to X$, government strategy is $\gamma^Y : H^t \times X \to Y$
- A pair of strategies (γ^X, γ^Y) is subgame perfect equilibrium (SPE) if
  - given behavior of future private agents and future governments, at each history government maximizes social welfare

$$\gamma^Y(h^{t-1}, x_t) \in \arg\max_{y \in Y} \left[r(x_t, y) + \sum_{s=0}^{\infty} \delta^{s+1} r(\gamma^X(\tilde{h}^{t+s}), \gamma^Y(\tilde{h}^{t+s})) \right] \quad \forall (h^{t-1}, x_t)$$

    where $\tilde{h}^t = (h^{t-1}, x_t, y)$, $\tilde{h}^{t+1} = (\tilde{h}^t, \gamma^H(\tilde{h}^t), \gamma^Y(\tilde{h}^t, \gamma^H(\tilde{h}^t)))$, etc.
  - private agents' expectations are rational

$$\gamma^H(h^{t-1}) = \gamma^Y(h^{t-1}, \gamma^H(h^{t-1}))$$

Recursive representation

- Problem of the government

$$\max_{y \in Y} \left[r(x_t, y) + \underbrace{\sum_{s=0}^{\infty} \delta^{s+1} r(\gamma^X(\tilde{h}^{t+s}), \gamma^Y(\tilde{h}^{t+s}))}_{\Gamma(\tilde{h}^t; \gamma^X, \gamma^Y) - \text{promised value}} \right] \quad \forall (h^{t-1}, x_t)$$

- How to rewrite it recursively? Make promised value, v_t, a state

$$\max_{y \in Y} (1 - \delta) \left[r(x_t, y) + \sigma_2(v_t, x_t, y) \right]$$

- Recursive government strategy is a pair of functions $\sigma = (\sigma_1, \sigma_2)$ together with the initial condition $v_1 \in \mathbb{R}$ such that

$$y_t = \sigma_1(v_t), \quad v_{t+1} = \sigma_2(v_t, x_t, y_t)$$

Subgame perfect equilibrium definition using recursive strategy

Recursive strategy σ with promised value v is subgame perfect equilibrium (SPE) iff

- $\sigma_2(v, \sigma_1(v), \eta)$ is value attainable by SPE $\forall \eta \in Y$
- promise keeping constraint is satisfied

$$v = (1 - \delta) r(y, y) + \delta \sigma_2(v, y, y)$$

  where $y = \sigma_1(v)$
- incentive compatibility constraint is satisfied

$$(1 - \delta) r(y, y) + \delta \sigma_2(v, y, y) \geq (1 - \delta) r(y, \eta) + \delta \sigma_2(v, y, \eta) \quad \forall \eta \in Y$$

- σ is both decision rule of the government and system of public expectations

Determining the set of all SPE values using APS

- Idea: We often look for equilibrium by iterating an operator which maps values into values. Here operators maps sets into sets.
- Start with the candidate set of equilibrium values $W \equiv [\underline{w}, \overline{w}] \subset \mathbb{R}$
- Which y can we implement?
- Pick $w_1 \in W$ as reward for adherence and $w_2 \in W$ as a punishment for deviation
- To relax incentive compatibility constraint as much as possible, set $w_2 = \underline{w}$

$$w = (1-\delta)r(y,y) + \delta w_1 \geq (1-\delta)r(y,\eta) + \delta \underline{w}, \quad \forall \eta \in Y$$

- Punishment is the same for any deviation, so government deviates to static best response

$$(1-\delta)r(y,y) + \delta w_1 \geq (1-\delta)r(y,B(y)) + \delta \underline{w}$$

More on APS

Algorithm

- Start with largest possible $W = [r(y^{\#}, y^{\#}), 0]$
- y is admissible if it satisfies IC for some $w_1 \in W$

$$(1-\delta)r(y,y) + \delta w_1 \geq (1-\delta)r(y, B(y)) + \delta \underline{w}$$

- For each admissible y define lowest and highest you can get

$$w_{LB}(y) = (1-\delta)r(y,y) + \delta \max\{\min_y w_1(y), \underline{w}\}$$
$$w_{UB}(y) = (1-\delta)r(y,y) + \delta \overline{w}$$

- Define new set as $W' = [\min_y w_{LB}(y), \max_y w_{UB}(y)]$
- Iterate until the set converges
- The result is the set of SPE values, V

Worst equilibrium

- Worst equilibrium solves

$$\underline{v} = \min_{y \in Y, v_1 \in V} \quad (1-\delta) r(y,y) + \delta v_1$$
$$\text{s.t.} \quad (1-\delta) r(y,y) + \delta v_1 \geq (1-\delta) r(y, B(y)) + \delta \underline{v}$$

- IC binds. If not, one can lower v_1 closer to $\underline{v}$.
- What if $v_1 = \underline{v}$ and you can't lower it (IC is slack)?
- Not possible. If $v_1 = \underline{v}$, then $\underline{v} = r(y^W, y^W)$ and IC gives $r(y^W, y^W) > r(y^W, B(y^W))$. Contradiction.
- IC and definition of the objective give

$$\underline{v} = (1-\delta) r(y, B(y)) + \delta \underline{v} \Rightarrow \underline{v} = r(y, B(y))$$

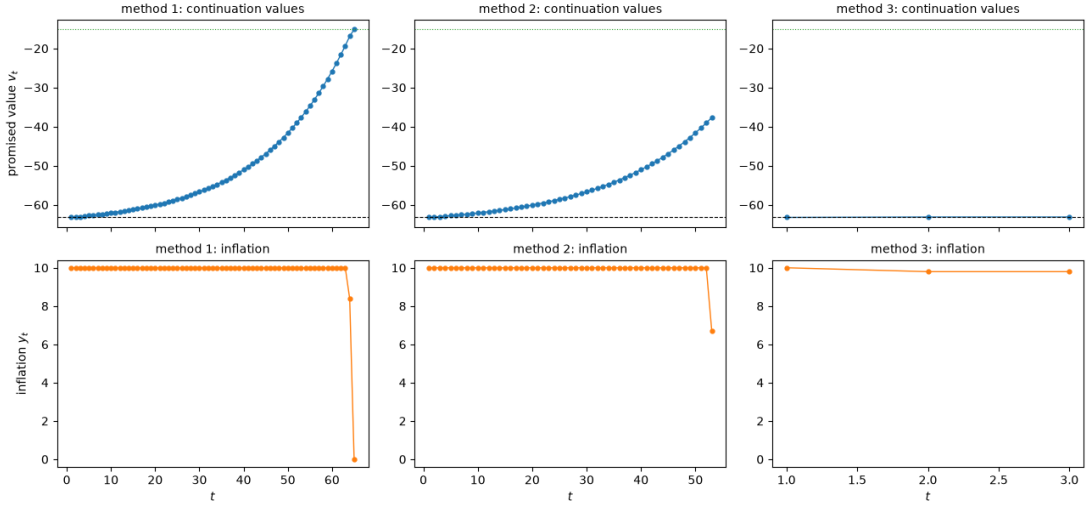
- Then

$$\underline{v} = \min_{\tilde{y} \in Y} r(\tilde{y}, B(\tilde{y})) = \min_{\tilde{y} \in Y} -\frac{(U^* + \theta \tilde{y})^2}{2(\theta^2 + 1)} = -\frac{(U^* + \theta y^\#)^2}{2(\theta^2 + 1)}$$

- Upper bound on inflation determines worst equilibrium

Examples of worst equilibria

Figures 4.2-4.4: three equilibria that all attain the worst value v_{min}



Best equilibrium

- Best equilibrium solves

$$\begin{aligned} \overline{v} = \max_{y \in Y, v_1 \in V} \quad & (1-\delta) r(y,y) + \delta v_1 \\ \text{s.t.} \quad & (1-\delta) r(y,y) + \delta v_1 \geq (1-\delta) r(y, B(y)) + \delta \underline{v} \end{aligned}$$

- $v_1 = \overline{v}$ (otherwise, not maximum)
- From the definition of the objective

$$\overline{v} = (1-\delta) r(y^B, y^B) + \delta \overline{v} \Rightarrow r(y^B, y^B) = \overline{v}$$

- So, the problem is

$$\begin{aligned} \overline{v} = \max_{y \in Y} \quad & r(y,y) \\ \text{s.t.} \quad & r(y,y) \geq (1-\delta) r(y, B(y)) + \delta \underline{v} \end{aligned}$$

- Best might be Ramsey outcome or something worse

Other examples of equilibria: infinite repetition of static Nash

- Don't use dynamic incentives: continuation is independent of current action
- Hence, government chooses static best response
- $\sigma_1(v) = y^N \quad \forall v$, $\sigma_2(v, x, y) = v^N \quad \forall (v, x, y)$

Infinite repetition of something better

- Let $v^b = r(y^b, y^b) > v^N$
- Supportable if IC is satisfied

$$\delta r(y^b, y^b) + (1-\delta)v^b \geq \delta r(y^b, B(y^b)) + (1-\delta)\underline{v}$$

- Profit from deviation should be lower than the loss in continuation value

$$\frac{\delta}{1-\delta}(v^b - \underline{v}) \geq r(y^b, B(y^b)) - r(y^b, y^b)$$

- Discount factor should be large enough
- Can replace worst punishment with infinite Nash if δ is large enough
- Nothing better than Ramsey can be sustained

Conclusion

- With infinite horizon one can sustain equilibria better than static Nash
- If agents are patient enough, we can sustain Ramsey outcome
- This is one way to think about credibility of the government
- One should be able to reproduce similar logic in popular NK model
- There are multiple equilibria in such an economy
- Issue of coordinating on a particular equilibrium with a particular reward-punishment schedule
- Same logic applies to other models with private agents and the government which feature time-inconsistency (capital taxation, patents, etc.)

Appendix

Abreu, Pearce, and Stacchetti

- Idea: We often look for equilibrium by iterating an operator which maps values into values. Here operators maps sets into sets.
- Start with the candidate set of equilibrium values $W \subset \mathbb{R}$
- Pick $w_1 \in W$ as reward for adherence and $w_2 \in W$ as a punishment for deviation
- (y, w) is admissible with respect to W if there exist $w_1, w_2 \in W$ with

$$w = (1-\delta)r(y,y) + \delta w_1 \geq (1-\delta)r(y,\eta) + \delta w_2, \quad \forall \eta \in Y$$

- Let $Z(W)$ collect all admissible pairs
- W is self-generating if $W \subseteq Z(W)$
- Set of SPE values V is the largest self-generating set
- Z maps compacts into compacts and is monotone, so one can find V using an iterative procedure

Abreu's stick and carrot

- Government inflates at $y^{\#}$ for one period
- Then rewarded with Ramsey forever

$$\tilde{v} = (1-\delta) r(y^{\#}, y^{\#}) + \delta v^{R}$$

- Use $\tilde{v}$ as punishment for refusing the stick
- Is incentive constraint satisfied?

$$\tilde{v} = \delta r(y^{\#}, y^{\#}) + (1-\delta) v^{R} \geq \delta r(y^{\#}, B(y^{\#})) + \delta \tilde{v}$$
$$\Rightarrow \tilde{v} \geq r(y^{\#}, B(y^{\#})) = \underline{v}$$